Ali NCIBI

Ph.D.
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cv

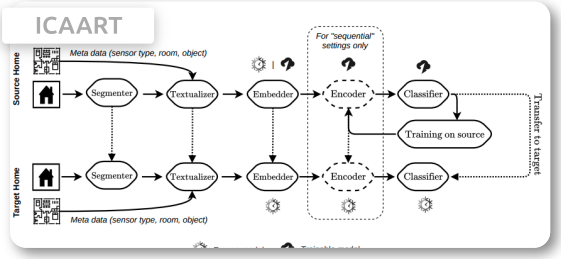
About Me

Researcher and engineer interested in machine learning, data, and intelligent systems. I enjoy building systems, exploring ideas, and turning complex problems into practical solutions.

Research Interests

- Machine Learning
- Human Activity Recognition
- Intelligent Systems
- Applied AI for real-world environments

Selected publications

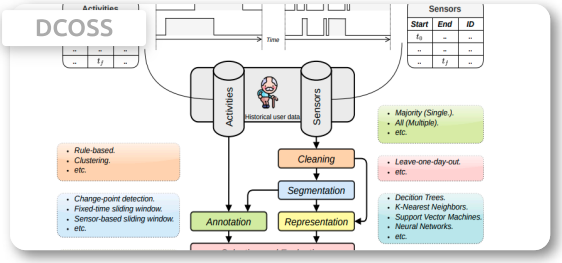


Towards Domain-Robust Activity Recognition using Textual Representations of Binary Sensor Events

Ali Ncibi

18th International Conference on Agents and Artificial Intelligence ***(ICAART)****, 2026.*

PDF Code BibTex

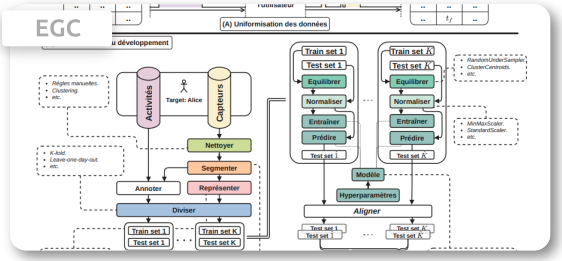


FlowAR: A Framework for Data-Driven Development of Human Activity Recognition Systems using Binary Sensors

Ali Ncibi, Luc Bouganim, Philippe Pucheral

*21st International Conference on Distributed Computing in Smart Systems and the Internet of Things **(DCOSS-IoT)**, 2025.*

PDF BibTex



FlowAR: une plateforme uniformisée pour la reconnaissance des activités humaines à partir de capteurs binaires

Ali Ncibi, Luc Bouganim, Philippe Pucheral

*25ème conférence francophone Extraction et Gestion des Connaissances **(EGC)**, 2025.*

PDF BibTex

Projects

FlowAR (Flows of Activity Recognition)

A modular, data-driven experimental framework designed to support reproducibility and scientific rigor in the development and evaluation of activity recognition systems for PhD research.

Tech: Scikit-learn, scikit-optim, Pytorch (Lightning), Scipy, Numpy, Pandas, Matplotlib, Plotly, Streamlit

Massirah CV Builder

A beautiful, responsive, multilingual web application that helps users create clean, ATS-optimized resumes, enhanced with AI-powered job matching and smart suggestions to improve and tailor CVs for specific roles.

Tech: Node.js, Express, PostgreSQL (via Supabase), React, Heroku

Live Demo

Flux Media Player

A media player and library organizer for personal streaming servers, enabling users to stream and manage IPTV content (Xtream and M3U) through a beautiful, responsive, and unified interface.

Tech: Node.js, Nuxt, Vercel

Live Demo

Temporal-links extraction between medical concepts from french medical text

Designed, implemented, and evaluated a deep learning–based approach for temporal relation (T-link) extraction from French clinical text corpora.

Tech: Pytorch

Style transfer between music accompaniments in MIDI

Applied deep-learning techniques to automatic music generation. Tweaked a generative model (Groove2Groove https://groove2groove.telecom-paris.fr) for the task.

Tech: Tensorflow

CNN-based registration of 2D brain tumor images

Studied a CNN-based approach for the task of brain tumor image registration. Evaluated different training losses, and explored the obtained transformations for examples from Brats MICCAI brain tumor dataset.

Tech: Tensorflow

Arrhythmia detection from ECG data

Designed and implemented an automated neural network for arrhythmia detection through the analysis of patients medical ECG recordings. Evaluated performance on examples from MIT-BIH Arrhythmia Database.

Tech: Tensorflow

Network standards for IoT apps. in healthcare

Reviewed and discussed networking trends for IoT in healthcare applications.

Tech: Tensorflow

Skills

Data Science Toolkit

- Supervised and unsupervised learning for classification and clustering tasks
- Descriptive statistics and statistical hypothesis testing
- Experience working with multiple data modalities including tabular, image, text, and audio data

Programming Languages

- Python
- JavaScript
- C/C++, C#
- SQL

Web & Mobile Development

- HTML5 and CSS3
- React
- Node.js
- Vue.js (Nuxt.js)
- Ionic / Capacitor

Tools

- Version control: Git
- Containerization: Docker
- Operating systems: Linux, macOS, Windows
- Cloud platforms: AWS, Google Cloud Platform (GCP), Heroku, DigitalOcean
- Productivity & authoring tools: Microsoft Office, Adobe Creative Suite, LaTeX

Education

PhD in Computer Science

Univeristy of Paris Saclay (UPS) — UVSQ/INRIA, Versailles/Saclay, France

2023 - 2026

Focus on the design of structured data-driven pipelines and the evaluation of semantic representation strategies for cross-home human activity recognition from IoT traces

MSc in Data Science

Telecom Paris, Institut Polytechnique de Paris (TPT/IPP) — Paris, France

2019 - 2022

Focus on numerical optimization, statistical machine learning, computer vision, natural language processing, graph mining, programming, and social & economic sciences.

MSc in Information Technology and Telecommunication

The National Engineering School of Tunis (ENIT) — Manar, Tunisia

2017 - 2019

Focus on signal transmission and processing, networking techniques, programming, databases, and system administration.

Preparatory class for the tunisian/french Grandes Ecoles systems

Institut préparatoire aux études scientifiques et techniques (IPEST) — Marsa, Tunisia

2014 - 2016

Specialized in mathematics and physics.